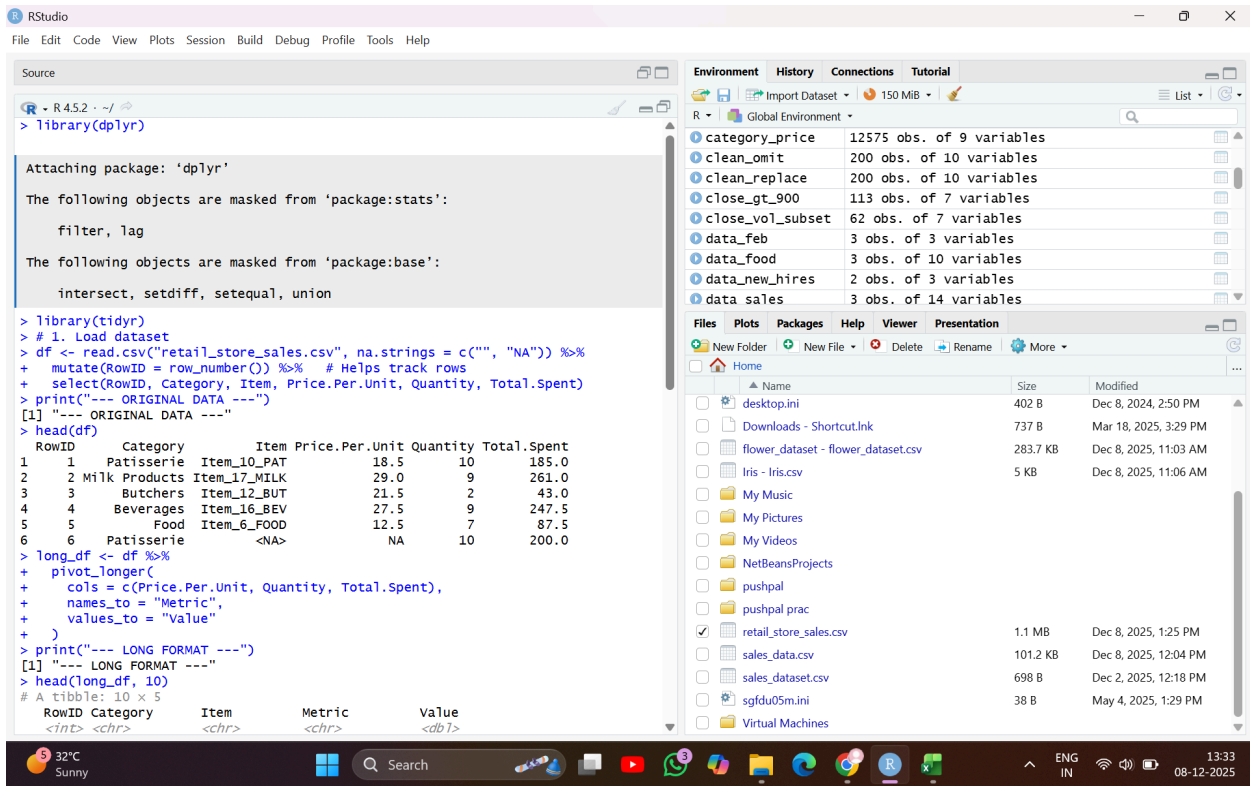


SHETH L.U.J AND SIR M.VCOLLEGE

Subject: Data Analysis with SAS / SPSS /R

Practical no. 11

Aim: Reshaping data using
`pivot_longer()/pivot_wider()` (R).



The screenshot displays the RStudio interface with the following components:

- Source Panel:** Contains R code for loading the `dplyr` and `tidyr` packages, reading a CSV file, and performing a `pivot_longer` operation. The code includes comments and a `print` statement to display the resulting long format data.
- Environment Panel:** Lists the loaded packages and their respective object counts and variable counts.
- Files Panel:** Shows the file explorer with various files and folders, including the `retail_store_sales.csv` file.

Code Snippet:

```
> library(dplyr)

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':
  filter, lag

The following objects are masked from 'package:base':
  intersect, setdiff, setequal, union

> library(tidyr)
> # 1. Load dataset
> df <- read.csv("retail_store_sales.csv", na.strings = c("", "NA")) %>%
+ mutate(RowID = row_number()) %>% # Helps track rows
+ select(RowID, Category, Item, Price.Per.Unit, Quantity, Total.Spent)
> print("--- ORIGINAL DATA ---")
[1] "--- ORIGINAL DATA ---"
> head(df)
  RowID Category Item Price.Per.Unit Quantity Total.Spent
1     1  Patisserie Item_10_PAT      18.5         10      185.0
2     2  Milk Products Item_17_MILK     29.0          9      261.0
3     3    Butchers Item_12_BUT      21.5          2       43.0
4     4  Beverages Item_16_BEV      27.5          9      247.5
5     5    Food Item_6_FOOD      12.5          7       87.5
6     6  Patisserie    <NA>         NA         10      200.0

> long_df <- df %>%
+ pivot_longer(
+   cols = c(Price.Per.Unit, Quantity, Total.Spent),
+   names_to = "Metric",
+   values_to = "Value"
+ )
> print("--- LONG FORMAT ---")
[1] "--- LONG FORMAT ---"
> head(long_df, 10)
# A tibble: 10 x 5
  RowID Category Item Metric Value
  <int> <chr>    <chr> <chr> <dbl>
```

Environment Panel Data:

Object	Obs.	Vars.
category_price	12575	9
clean_omit	200	10
clean_replace	200	10
close_gt_900	113	7
close_vol_subset	62	7
data_feb	3	3
data_food	3	10
data_new_hires	2	3
data_sales	3	14

Files Panel Data:

Name	Size	Modified
desktop.ini	402 B	Dec 8, 2024, 2:50 PM
Downloads - ShortcutLink	737 B	Mar 18, 2025, 3:29 PM
flower_dataset - flower_dataset.csv	283.7 KB	Dec 8, 2025, 11:03 AM
Iris - Iris.csv	5 KB	Dec 8, 2025, 11:06 AM
My Music		
My Pictures		
My Videos		
NetBeansProjects		
pushpal		
pushpal prac		
retail_store_sales.csv	1.1 MB	Dec 8, 2025, 1:25 PM
sales_data.csv	101.2 KB	Dec 8, 2025, 12:04 PM
sales_dataset.csv	698 B	Dec 2, 2025, 12:18 PM
sgfdu05m.ini	38 B	May 4, 2025, 1:29 PM
Virtual Machines		

